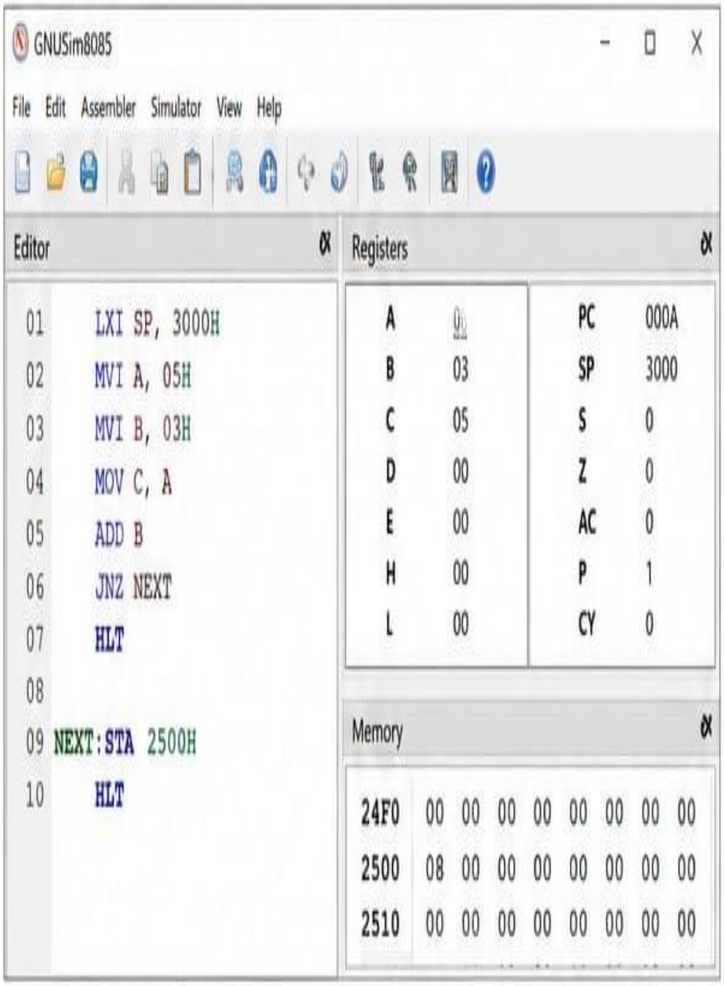
CO1 – ASSESSMENT TOOL 1

Q1. 8085 PROGRAM EXECUTION IN GNUSIM8085

Program: Data transfer, arithmetic and branch instructions



Observation:

- After execution, A = 08H
- Result 08H is stored in memory location 2500H
- PC points to the last HLT instruction
- SP remains 3000H

Explanation:

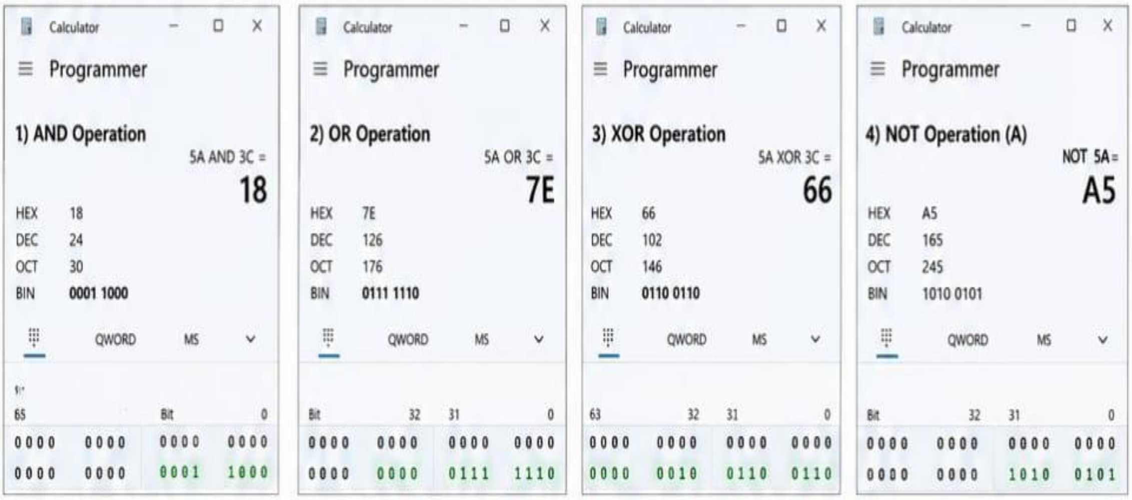
1. MVI A,05H → Load 05H into A
2. MVI B,03H → Load 03H into B
3. MOV C,A → Copy contents of A to C
4. ADD B → Add B to A (05H + 03H = 08H)
5. JNZ NEXT → Since Zero flag = 0, jump to NEXT
6. STA 2500H → Store A (08H) at memory 2500H
7. HLT → Stop execution

Fetch – Decode – Execute Cycle

Cycle	Description	Example (ADD B)
Fetch	CPU fetches the instruction from memory using PC	PC points to ADD B
Decode	Control Unit decodes the instruction	Instruction identified as ADD
Execute	ALU performs addition, result stored in A	A = 08H, Flags updated
Update	PC updated to next instruction	PC = address of next instruction

Q2. BITWISE OPERATIONS USING WINDOWS CALCULATOR (PROGRAMMER MODE)

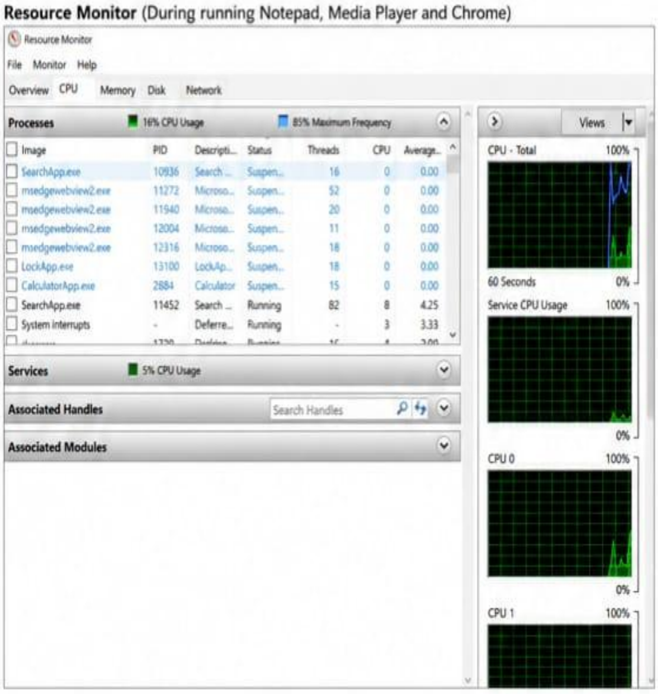
Values taken: A = 5AH B = 3CH



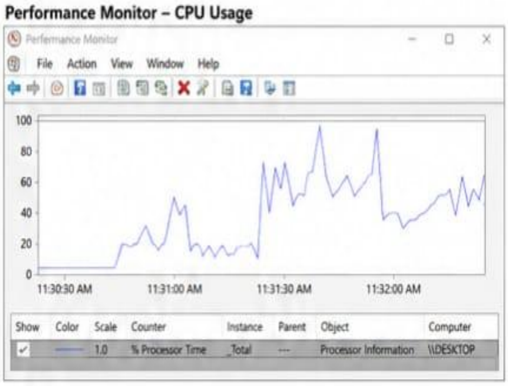
Analysis: Bitwise operations manipulate individual bits of data. These operations are widely used in masking, toggling, data comparison, encryption, and hardware control in computer systems.

Q3. OBSERVATION USING RESOURCE MONITOR AND PERFORMANCE MONITOR

Resource Monitor (During running Notepad, Media Player and Chrome)



Performance Monitor – CPU Usage



Analysis:

- CPU is the main processing unit that executes instructions of Notepad, Media Player and Chrome.
- Memory stores program code and data temporarily for quick access.
- Disk performs read/write operations to load programs and save files.
- Network handles internet operations for the browser.
- I/O devices (keyboard, mouse, display, speakers) provide input and output to the user.
- All these functional units work together to complete multitasking efficiently.